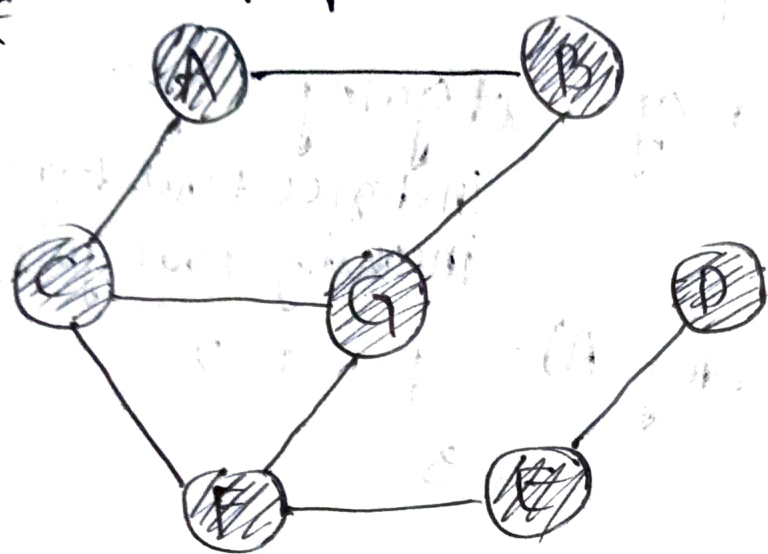


DFS(B)
DFS(A)
call stack
H[E]
adjace

graph-2

*



visited:

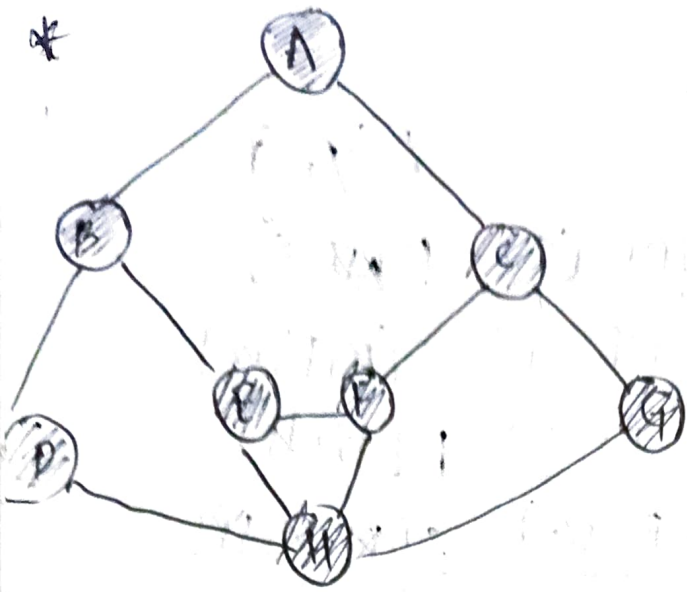
A	B	C	D	E	F	G
✓	✓	✓	✓	✓	✓	✓
1	1	1	1	1	1	1

output: A B G C F E D

DFS(D)
DFS(E)
DFS(F)
DFS(G)
DFS(B)
DFS(A)
call stack

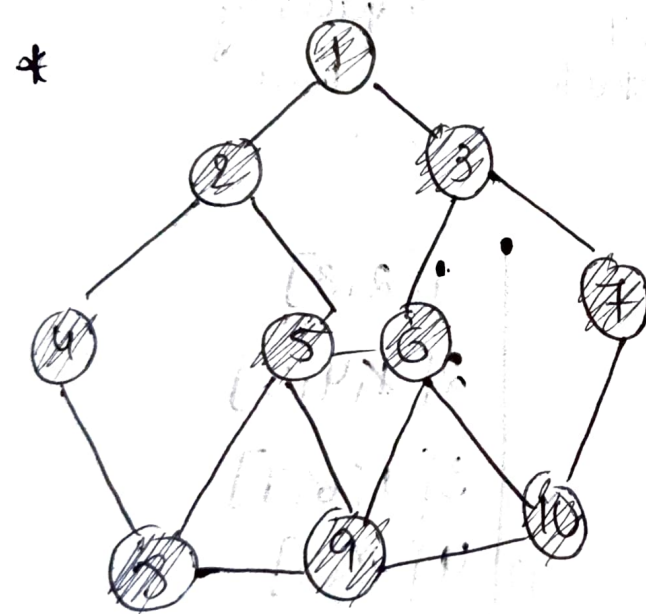
A[B, C]
B[A, G]
C[A, F, G]
D[]
E[D, F]
F[C, E, G]
G[B, F, E]

adjacency list



DFS(G)	A [B, C]
DFS(B)	B [A, D, E]
DFS(C)	C [A, F, G]
DFS(E)	D [B, H]
DFS(H)	E [B, F, H]
DFS(D)	F [E, H]
DFS(F)	G [E, H]
DFS(A)	H [D, E, F, G]

output: A B D H E F G



DFS(9)	1 [2, 3]
DFS(10)	2 [1, 4, 5]
DFS(7)	3 [1, 6, 7]
DFS(3)	4 [2, 8]
DFS(6)	5 [2, 6, 8, 9]
DFS(5)	6 [3, 5, 9, 10]
DFS(8)	7 [3, 10]
DFS(4)	8 [4, 5, 9]
DFS(2)	9 [5, 6, 8, 10]
DFS(1)	10 [6, 7, 9]

output: 1, 2, 4, 8, 5, 6, 3, 7, 10, 9

graph-2 adjacency matrix:

	A	B	C	D	E	F	G
A	0	1	1	0	0	0	0
B	1	0	0	0	0	0	1
C	1	0	0	0	0	1	1
D	0	0	0	0	1	0	0
E	0	0	0	1	0	1	0
F	0	0	1	0	1	0	1
G	0	0	1	0	1	1	0

A	B	C	G
B	A	G	
C	A	F	G
D	E		
E	D	F	
F	C	E	G
G	C	F	G

graph-3:

	A	B	C	D	E	F	G	H
A	0	1	1	0	0	0	0	0
B	1	0	0	1	1	0	0	0
C	1	0	0	0	1	1	0	0
D	0	1	0	0	0	0	0	1
E	0	1	0	0	0	1	0	1
F	0	0	1	0	1	0	0	1
G	0	0	1	0	0	0	0	1
H	0	0	0	1	1	1	1	0

A	B	C
B	A	D
C	A	F
D	B	H
E	B	F
F	C	E
G	C	H
H	D	E

graph-4

	1	2	3	4	5	6	7	8	9	10
1	0	1	1	0	0	0	0	0	0	0
2	1	0	0	1	1	0	0	0	0	0
3	1	0	0	0	0	1	1	0	0	0
4	0	1	0	0	0	0	0	1	0	0
5	0	1	0	0	0	1	0	1	1	0
6	0	0	1	0	1	0	0	0	1	1
7	0	0	1	0	0	0	0	0	0	1
8	0	0	0	1	1	0	0	0	1	0
9	0	0	0	0	1	1	0	1	0	1
10	0	0	0	0	1	1	0	0	1	0

1	2	3
2	1	4
3	1	6
4	2	3
5	2	6
6	3	5
7	3	4
8	4	5
9	5	6
10	6	7